Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 1.0 | **Date:** 07-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A 12-month Open-label, Phase 4 Multicenter Safety Study to Evaluate the Effect of KarXT on Voiding Dynamics and Urological Safety **Short Title/Acronym:** KarXT Urological Safety Study **Protocol Number:** CN012-0068 **NCT Number:** NCT07221877 **Posting ID:** 426698389078224667 **Trial Status:** Not Yet Recruiting (Not yet recruiting participants) **Sponsor/Company Name:** Bristol Myers Squibb **Investigational Product:** KarXT (xanomeline/trospium) (Trospium Trade Name: Sanctura) **Phase of Development:** Phase 4 (PHASE4) **Medical Monitor:** BMS Clinical Trials Contact Center

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2. Study Synopsis & Rationale

2.1 Study Objective Summary/Primary Objective: The purpose of this study is to characterize the effect of KarXT on voiding dynamics and urological safety in participants with DSM-5 schizophrenia. **Secondary Objectives:** To evaluate the broader long-term safety, tolerability, and clinical response of KarXT within the targeted patient population.

2.2 Background & Rationale To systematically characterize the effect of KarXT on voiding dynamics and urological safety in participants with DSM-5 schizophrenia. Given the cholinergic and anticholinergic mechanisms of the combined xanomeline/trospium formulation, it is a medical necessity to robustly evaluate potential urological safety impacts, including urine retention or flow alterations, in an outpatient real-world equivalent population.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target N: 60 evaluable patients **Number of Sites:** Multicenter (Multiple US sites) **Estimated Study Start:** Q4 2024 **Estimated Primary Completion:** Dec-2027 (Overall span of approximately 102 weeks)

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and Female participants, 18 to 65 years of age, with a primary diagnosis of schizophrenia based on psychiatric evaluation (DSM-5) and confirmed by the Mini-

International Neuropsychiatric Interview (MINI v 7.0.2). 2. Participants must have a PANSS total score $\leq$ 80 and CGI-S score $\leq$ 4, at screening and baseline, and a BMI $\geq$ 18 and $\leq$ 40 kg/m2. 3. Participants must be willing and able to discontinue all antipsychotic medications prior to the baseline visit and be willing and able to comply with protocol requirements.

4.2 Exclusion Criteria 1. Participants with newly diagnosed schizophrenia, any other DSM-5 disorder diagnosed within the past 12 months, alcohol or drug use disorder within the past 12 months, or history/presence of clinically significant disease or disorder that would jeopardize participant safety or validity of study results. 2. Participants at risk for suicidal behavior, as well as individuals who are pregnant or breastfeeding, will be excluded from the study. 3. Inability or refusal to comply with urological testing (e.g., uroflowmeter usage). 4. Other protocol-defined Inclusion/Exclusion criteria apply.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: KarXT (xanomeline/trospium) administered as prescribed per standard dosing guidelines for schizophrenia. **Route of Administration:** Oral **Duration of Treatment:** Up to 12 months (overall study framework up to 102 weeks).

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of permitted background psychotropic medications. **Prohibited:** Concomitant use of systemic medications with strong overlapping anticholinergic or cholinergic profiles that would heavily confound urological safety readouts.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Psychiatric Evaluation (MINI v 7.0.2), Baseline Urological Symptom Assessment
Baseline	Day 0	First Dose, Baseline Uroflowmetry and PVR measurements
Interim	Scheduled periodic	Safety Labs, Efficacy Assessment, Flowstar Uroflowmetry & PVR testing
End of Study	Week 102	Final Urological Assessment, Exit Interview, IP Accountability

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7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. Change from baseline in voiding dynamics (Qmax) over time. 2. Change from baseline in post-void residual volume (PVR) over time. *(Measurement Method Context: Qmax tracked via Flowstar Compact Digital Uroflowmeter; PVR measured to assess urine left in the bladder immediately after urination).*

7.2 Secondary & Exploratory Outcomes 1. Number of participants with a change from baseline in urinalysis parameters over time. 2. Change from baseline in IPSS over time. 3. Number of participants with electrocardiogram (ECG) abnormalities. 4. Incidence and severity of general Adverse Events (AEs) and Serious Adverse Events (SAEs). 5. General tolerability of KarXT therapy over the longitudinal follow-up period.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via site visits, focusing specifically on urological symptoms, urinary retention, hesitancy, and any "Adverse event affecting urinary system" (Preferred UMLS Name). **Serious Adverse Events (SAEs):** Standard reporting timeline within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Supervised by internal BMS safety review committees.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants receiving at least one dose of KarXT). **Sample Size Justification:** Targeted Enrollment = 60, sized sufficiently to detect meaningful signals or deviations in urological safety parameters (PVR and flow metrics). **Handling of Missing Data:** Last Observation Carried Forward (LOCF) or Mixed-Model Repeated Measures (MMRM) for longitudinal voiding dynamic trends.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring of investigator sites to ensure protocol adherence, specifically regarding the correct use of the Flowstar Compact Digital Uroflowmeter. **Audit Trail:** Standardized electronic Case Report Form (eCRF) data entry and device-generated printouts utilized as source data verification.

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11. Study Identifiers & Governance

Primary ID: CN012-0068 **Posting ID:** 426698389078224667 **Registry Number:** NCT07221877 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study to Evaluate the Effect of KarXT on Urological Safety **Official Regulatory Title:** A 12-month Open-label, Phase 4 Multicenter Safety Study to Evaluate the Effect of KarXT on Voiding Dynamics and Urological Safety **Drug ATC Code:**

G04BD09 **Condition MeSH Code**: D012559 **Condition SNOMED ID**: 58214004 **Condition HPO Code**: HP:0100753

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12. Clinical Framework (Schizophrenia Specific)

Primary Condition / Disease Area: Schizophrenia **Semantic Type:** Mental or Behavioral Dysfunction **Disease Definition:** A major psychotic disorder characterized by abnormalities in the perception or expression of reality. It affects the cognitive and psychomotor functions. Common clinical signs and symptoms include delusions, hallucinations, disorganized thinking, and retreat from reality. **Diagnosis Threshold:** DSM-5 criteria confirmed by MINI (v 7.0.2). Condition maps to SNOMED Hierarchy: 69322001 (Psychotic disorder) -> 74732009 (Mental disorder) -> 64572001 (Disease). **Medical Methodology:** Pharmacological management with Dual M1/M4 Muscarinic Agonist (KarXT / trospium). **Optimization Target:** Maintaining effective psychiatric management while demonstrating favorable urological safety (normal PVR and voiding flow parameters).

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13. Study Procedures & Methodology

Management Pathway: Urological Safety Tracking during active Schizophrenia treatment **Education Protocol (HCP):** Site staff training on correct administration of the Flowstar Compact Digital Uroflowmeter and ultrasound PVR measurements. **Education Protocol (Patient):** Instructions for standardized voiding inside clinic settings (funnel or commode apparatus usage). **Quality Audit Mechanism:** Real-time printout data capture from the digital uroflowmeter and device calibration logs.

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14. Observation & Follow-Up Schedule

Total Duration: ~102 weeks of span (12-month primary intervention period) **Onsite Visit Cadence:** Regular intervals aligned with standard Phase 4 safety tracking for voiding testing. **Remote Monitoring Frequency:** Periodic tele-health/phone follow-ups as appropriate for safety reporting. **Checklist Review Interval:** Standardized questionnaire reviews for urinary retention or discomfort at each study visit.

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	**[DD-MMM-YYYY]**
Clinical Operations Lead	**[Name]**	___	***[DD-MMM-YYYY]***
Lead Statistician	***[Name]***	_____	[DD-MMM-YYYY]